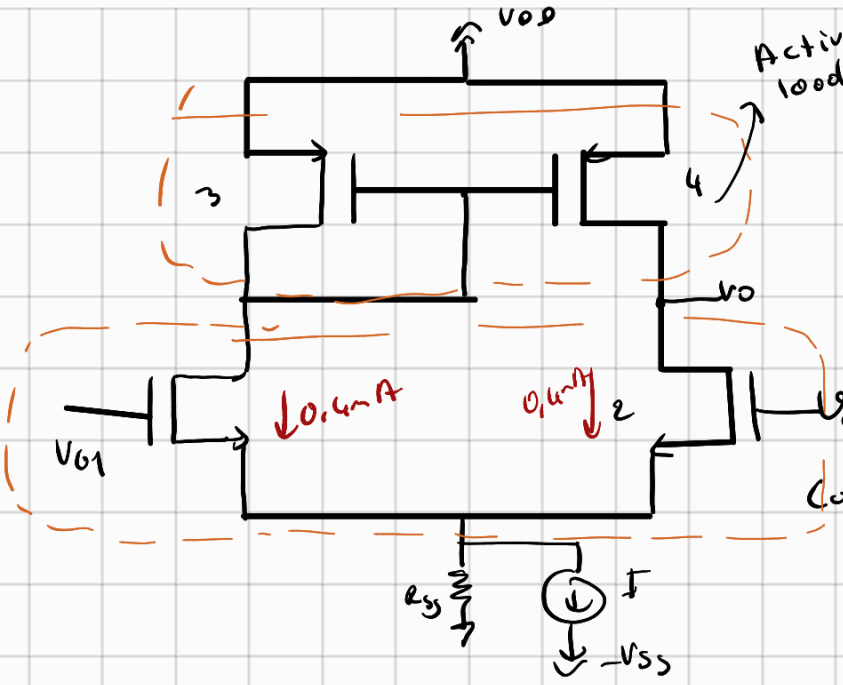
(7)



Consider the active load MOS differential amplifier with $\left(\frac{W}{L}\right)_n = 100$, $\left(\frac{W}{L}\right)_p$

$\mu_n C_{ox} = 2\mu_p C_{ox} = 0,2 mA/V^2$

$|V_A|_n = |V_A|_p = 20V$

$I = 0,8 mA$, $R_{SS} = 25k\Omega$

Calculate g_m, R_o, $|A_d|$, $|A_c|$ and CMRR

$g_{m1,2} = ?$ $\rightarrow \frac{2(0,4)}{V_{OV}=0,2} = 4mS$

$R_{SS} = 25k\Omega$

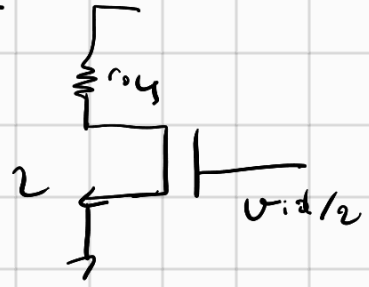
$r_{o_{1,2}} = \frac{V_A}{I_D} = \frac{20}{0,4} = 50k\Omega$

$0,4 = \frac{1}{2} 0,2 \cdot 100 (V_{OV})^2$

$0,04 = (V_{OV})^2 \Rightarrow V_{OV} = 0,2$

$r_{o_{3,4}} = r_{o_{1,2}} = 50k\Omega$

differential mode



$= g_m (r_{o4} \| r_{o2})$

$= \frac{1}{2} g_m r_o = \frac{1}{2} \cdot 4mS \cdot 50k\Omega$

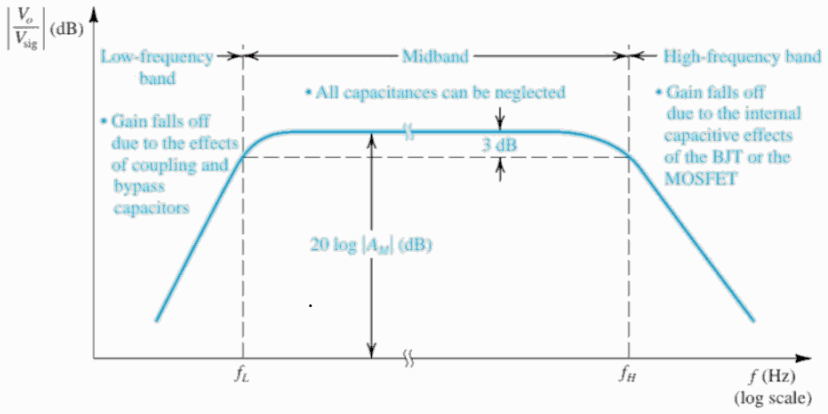
$= 100 [V/V]$

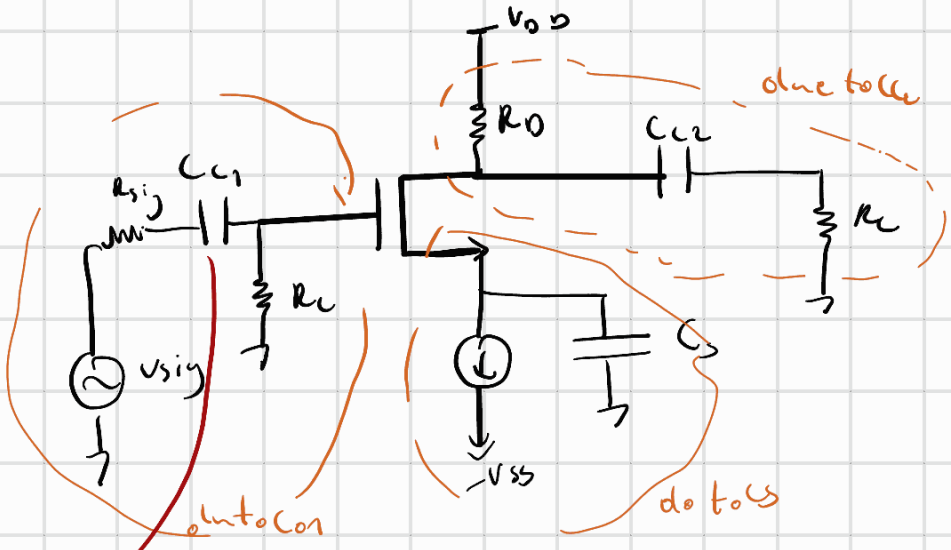
$A_{cm} = \frac{1}{2 g_m R_{SS}} = \frac{1}{2 \times 4mS \times 25k\Omega} = \frac{1}{200}$

$CMRR = \frac{100}{\frac{1}{200}} = 20\,000$

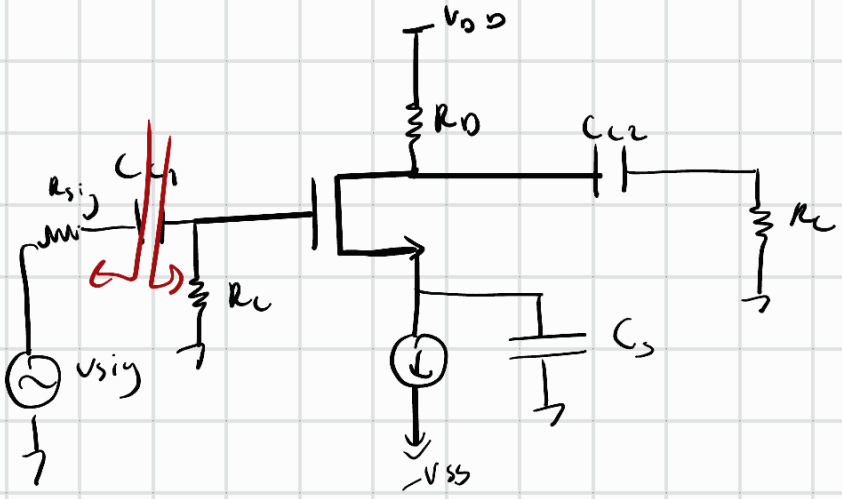
$20 \log (20 \cdot 10^3) = 86 dB$

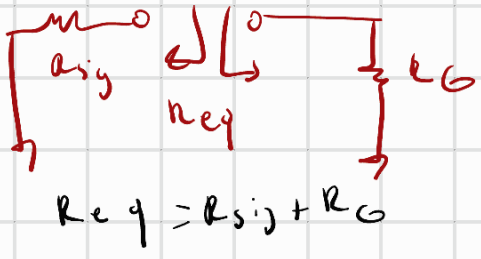
Frequency response





$$\frac{1}{j\omega C}$$

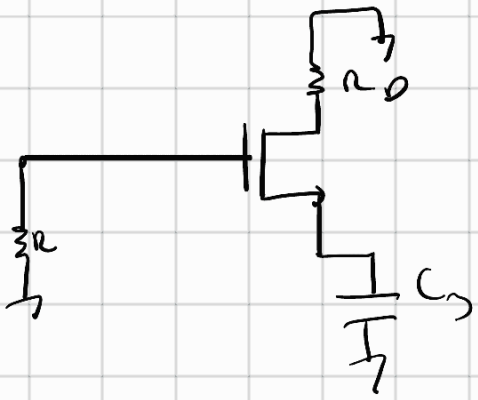


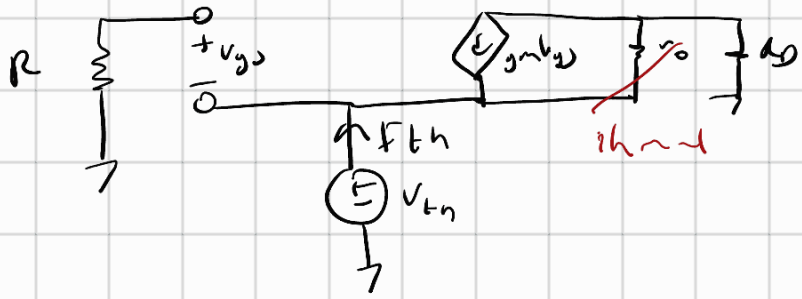


$$R_{eq} = R_{sig} + R_G$$

$$\omega_{P1} = \frac{1}{C_{C1}(R_{sig} + R_{EE})}$$

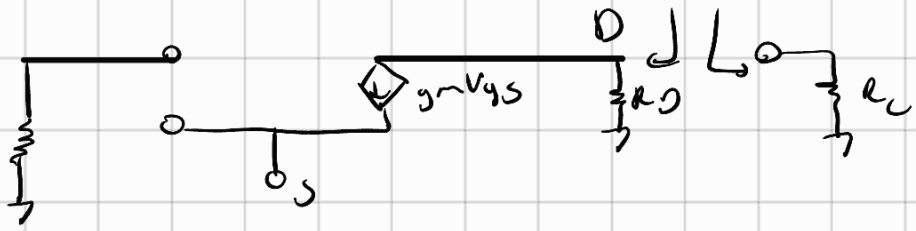
2. kapasiteden kaynaklı ethigi kutup



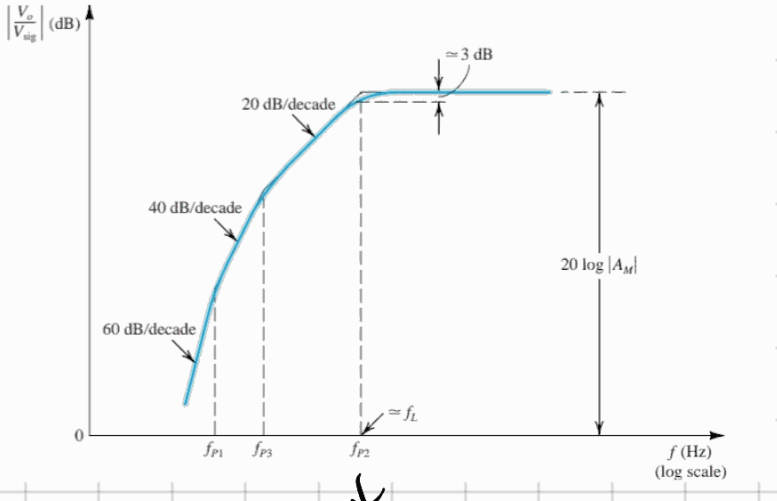


$$\frac{V_{th}}{I_{th}} = ? = \frac{-V_{gs}}{-g_m V_{gs}} = \frac{1}{g_m}$$

$$\omega_{p_2} = \frac{1}{\frac{C_S}{g_m}} = \frac{g_m}{C_S}$$



$$\omega_{p_1} = \frac{1}{(R_D + R_L) C_{L2}}$$



↓ 2 pole

sınırlı diferansiyel

& dominant pole

diğerlerinin

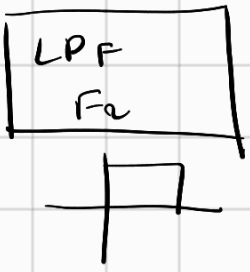
Shunt yaklaşım

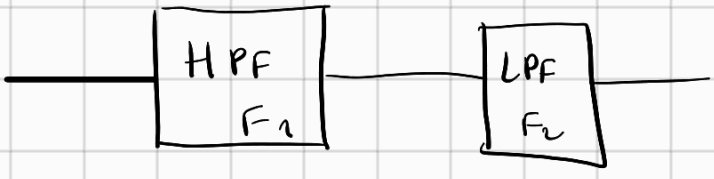
$$A_{mid} = -\left[\frac{R_G}{R_{sig}+R_G}\right] g_m (R_D \| R_L)$$

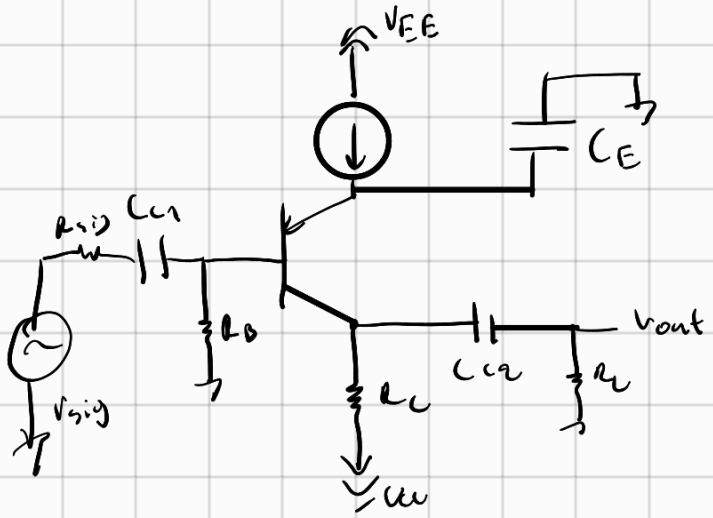
$$A(s) = A_{mid}\left[\frac{s}{s+\omega_{P1}}\right]\left[\frac{s}{s+\omega_{P2}}\right]\left[\frac{s}{s+\omega_{P3}}\right]$$

$$f_L = \frac{1}{2\pi}\left(\frac{1}{R_1 C_1} + \frac{1}{R_2 C_2} + \frac{1}{R_3 C_3}\right)$$

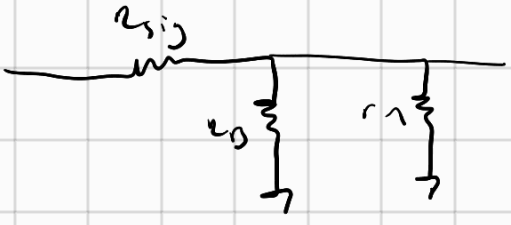




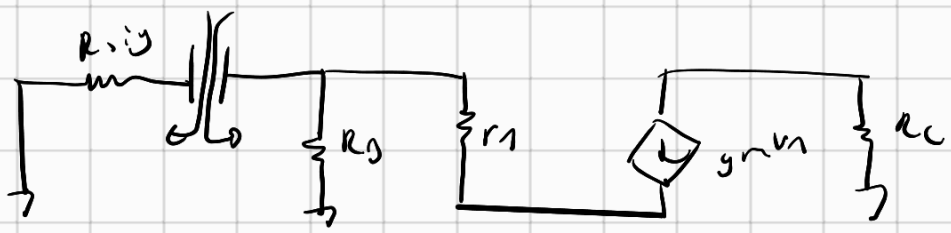


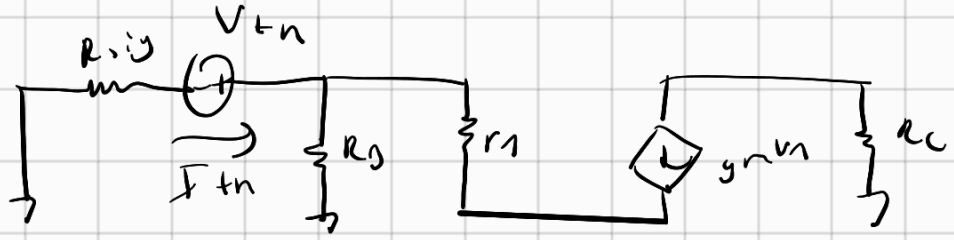


$$A_m = -g_m (R_D \| R_L) \frac{(R_G \| r_\pi)}{R_{sig} + (R_G \| r_\pi)}$$



$$A_m = -g_m (R_D \| R_L) \frac{(R_B \| r_\pi)}{R_{sig} + (R_B \| r_\pi)}$$

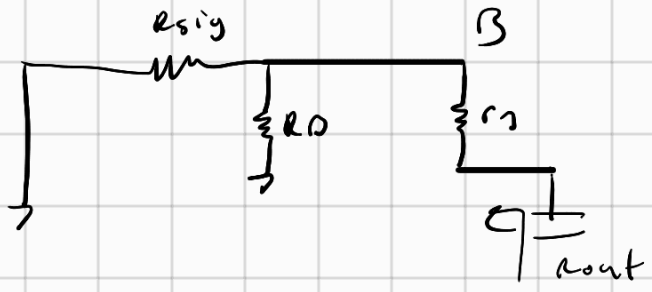




$$R_1 = R_{CC_1} = \left(R_{sig} + (R_D \,//\, r_1)\right)$$

$$w_{P1} = \frac{1}{C_{C_1}\left(R_{sig} + (R_D \,//\, r_1)\right)}$$

C_E den bakarken



Low frekansta C'ler short kabul edilir

$$R_{CE} = R_2 = \frac{(R_{sig} \,//\, R_D) + r_1}{\beta + 1}$$

$$w_{P2} = \frac{1}{\frac{(R_{sig} \,//\, R_B) + r_1}{\beta + 1}\, C_E}$$

$$w_{P3} = \frac{1}{C_{C_2}(R_C + R_{out})}$$